

Problem 18.17

Which of the following are characteristics of all perpetuities, including non-level payments?

- I. The present value is equal to the first payment divided by the annual effective interest rate.
- II. Payments continue forever.
- III. Each payment is equal to the interest earned on the principal.

Solution

We check each statement.

Statement I

For a level perpetuity-immediate with payment X , the present value is

$$PV = \frac{X}{i}.$$

However, the statement asks about all perpetuities, including non-level payments. For example, in a growing perpetuity with first payment X and growth rate g , the present value is

$$PV = \frac{X}{i - g},$$

not necessarily X/i . Therefore, statement I is false.

Statement II

By definition, a perpetuity is a stream of payments that continues forever. Therefore, statement II is true.

Statement III

For a level perpetuity, the payment may equal the interest earned on the principal. However, this need not be true for a non-level perpetuity. For example, if payments grow over time, later payments will not all equal the same interest earned on a fixed principal. Therefore, statement III is false.

Hence, the only statement that is true for all perpetuities is

II only.